

**TED(15) - 3241
(REVISION-2015)**

MODEL

**THIRD SEMESTER DIPLOMA EXAMINATION IN BIOMEDICAL ENGINEERING
BASIC MEDICAL SCIENCE**

(Maximum Marks : 100)

TIME : 3HRS

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

- 1) Define tissue?
- 2) List two cold applications in medicine?
- 3) State "expiratory reserve volume"?
- 4) Write the role of enzymes in digestion?
- 5) Name any four receptors present in human body?

5x2=10 marks

PART - B

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

- 1) Differentiate between monophasic and biphasic recording?
- 2) Describe different types of EMG electrode?
- 3) Explain electromyography?
- 4) Describe the general organization of brain?
- 5) Comprehend the physiology of respiration?
- 6) Draw and mark the parts of kidney?
- 7) Explain the role of liver in digestion?

5x6=30 marks

PART - C

Answer one full question from each Unit. Each question carries 15 marks.

UNIT I

- III** a) With a neat sketch explain the role of rod cells? 8
b) Explain different types of epithelial tissues? 7

OR

- IV.** a) Sketch the wave form of action potential and explain its different phases ? 8
b) Explain the mechanism of action of osmoreceptors? 7

UNIT II

- V.** a) Describe the temperature regulation in human body? 9
b) Classify different types of cardiac myopathies? 6

OR

- VI** a) Write a short note on heat application in medicine? 6
b) Describe cardio vascular system? 9

UNIT III

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| VII a) Draw and label the parts of respiratory system? | 8 |
| b) Explain the role of kidney in maintenance of water balance? | 7 |

OR

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| VIII a) Summarize the mechanism of formation of urine? | 8 |
| b) Explain the different phases of respiration? | 7 |

UNIT IV

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| IX a) Describe the digestion of fat? | 8 |
| b) Distinguish the absorption process of digested food products? | 7 |

OR

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| X a) Outline the mechanism of equilibrium balance? | 8 |
| b) Explain the role of vitamin A in photochemistry of vision? | 7 |

15x4=60